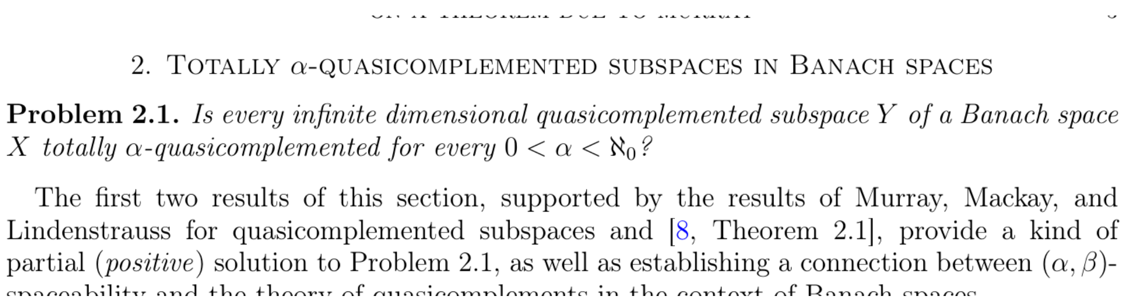
A finite-codimensional counterexample to Problem 2.1

arXiv:2403.05806, “On a theorem due to Murray”

Source Problem

Problem 2.1 of Barbosa, Raposo Jr., and Ribeiro asks whether every infinite-dimensional quasi-complemented subspace Y of a Banach space X is totally α-quasicomplemented for every finite $0 < \alpha < \aleph_0$ [1].



2. Totally α-quasicomplemented subspaces in Banach spaces

Problem 2.1. *Is every infinite dimensional quasicomplemented subspace Y of a Banach space X totally α-quasicomplemented for every $0 < \alpha < \aleph_0$?*

The first two results of this section, supported by the results of Murray, Mackay, and Lindenstrauss for quasicomplemented subspaces and [8, Theorem 2.1], provide a kind of partial (*positive*) solution to Problem 2.1, as well as establishing a connection between (α, β)-

Figure 1: Problem 2.1 in the source paper.

Definitions Used

Following the source paper, a closed subspace $Y \subset X$ is quasicomplemented if there is a closed subspace $Z \subset X$ such that $Y \cap Z = \{0\}$ and $Y + Z$ is dense in X. It is totally α-quasicomplemented if Y is quasicomplemented and, for every α-dimensional subspace $F \subset X$ with $F \cap Y = \{0\}$, there is a quasicomplement Z of Y containing F such that $\dim Z/F = \infty$.

Proof Intuition

The definition of total α-quasicomplementation demands room beyond the prescribed finite-dimensional subspace F: the eventual quasicomplement Z must contain F and still have infinite-dimensional quotient Z/F. If Y has finite codimension and F already exhausts the quotient X/Y, then no such room exists. Any subspace disjoint from Y and containing F is forced to be just F.

Counterexample

Theorem 1. *Problem 2.1 in [1], as stated, has a negative answer.*

Proof. Let

$$X = \ell_2 \oplus_2 \mathbb{R}, \qquad Y = \ell_2 \oplus \{0\}, \qquad F = \{0\} \oplus \mathbb{R}.$$

Then X is a Banach space and Y is a closed infinite-dimensional subspace. Moreover, Y is quasi-complemented: F is closed, $Y \cap F = \{0\}$, and $Y + F = X$.

We show that Y is not totally 1-quasicomplemented. Let Z be any quasicomplement of Y in X with $F \subset Z$. Since

$$X = Y \oplus F,$$

every $z \in Z$ can be written uniquely as $z = y + f$, with $y \in Y$ and $f \in F$. Because $f \in F \subset Z$, we have

$$y = z - f \in Z \cap Y = \{0\}.$$

Thus $z = f \in F$, and so $Z \subset F$. Since $F \subset Z$, it follows that $Z = F$. Consequently,

$$\dim Z/F = 0,$$

not infinity. Therefore the required condition in the definition of total 1-quasicomplementation fails for this one-dimensional subspace F.

Hence an infinite-dimensional quasicomplemented subspace need not be totally α-quasicomplemented for every finite $0 < \alpha < \aleph_0$. □

Verification and Scope

No computational verification is needed. The counterexample only uses the direct-sum decomposition of a codimension-one closed subspace. It answers Problem 2.1 exactly as stated. It does not challenge the paper's later positive results for infinite-codimensional subspaces.

Novelty Check

The run indexes were searched for `2403.05806`, `Murray`, `quasicomplement`, `totally alpha`, `hyperplane`, and `finite codimension`; no prior local packet or attempt was found. Web searches for the exact problem phrase, the paper title together with `finite codimension`, and the arXiv id together with `quasicomplemented Problem` did not reveal a separate known answer. This is a bounded novelty check rather than an exhaustive literature survey.

References

[1] Anderson Barbosa, Anselmo Raposo Jr., and Geivison Ribeiro, *On a theorem due to Murray*, arXiv:2403.05806, 2024. `https://arxiv.org/abs/2403.05806`

Human Review: Finite-codimensional quasicomplement counterexample

Original automatic proof: `solution_packet.pdf`
Checked date: 18 June 2026
Status: Manually checked.
Verdict: The proof appears correct as a literal counterexample.

Human Comments

The proposed solution appears correct as an answer to Problem 2.1 exactly as stated in Barbosa–Raposo–Ribeiro's paper. The example is

$$X = \ell_2 \oplus_2 \mathbb{R}, \qquad Y = \ell_2 \oplus \{0\}, \qquad F = \{0\} \oplus \mathbb{R}.$$

Then Y is closed, infinite-dimensional, and quasicomplemented. In fact, it is 1-complemented, so the example is very elementary.

The obstruction is finite codimension. Since

$$X = Y \oplus F,$$

any subspace Z which is disjoint from Y and contains F must equal F. Indeed, if $z \in Z$, write $z = y + f$ with $y \in Y$ and $f \in F$. Since $F \subset Z$, one has $y = z - f \in Z \cap Y = \{0\}$, so $z = f \in F$. Hence $Z = F$, and therefore

$$\dim Z/F = 0,$$

not infinity.

Interpretation

Mathematically, the argument is correct. However, this seems likely to be a minor issue in the formulation of the question rather than a substantial counterexample to the intended problem. The example is simply a finite-codimensional complemented subspace, and the obstruction comes from the fact that the chosen finite-dimensional F already exhausts the quotient X/Y. There is then no room left for a quasicomplement Z containing F with infinite-dimensional quotient Z/F.

This strongly suggests that the intended question was probably meant in the setting of infinite codimension, which is also the setting emphasized by the paper's later positive results.

Literature Check

A preliminary online check did not find a later answer to this exact point. OpenAlex reports no citing works for arXiv:2403.05806, a direct OpenAlex citing-works query returned no results, and OpenCitations returned an empty citation list for the arXiv DOI. Targeted searches for the terminology around total α-quasicomplementation, finite codimension, and Problem 2.1 did not reveal another source discussing this example.

Review Outcome

The packet should be recorded as a correct literal counterexample, but with the caveat that it probably identifies a missing hypothesis or intended restriction in the question. I would not present it as a deep mathematical contribution; rather, it is a semantic warning about finite-codimensional cases.